Le Chatelier's Principle

Answer key with worked solutions

A. Understanding

1. Le Chatelier’s principle.

> If a system at equilibrium is disturbed by a change in concentration, pressure or temperature, the position of equilibrium shifts in the direction that **partly counteracts** the change.

B. The Haber process

2. Predictions.

> (a) Shifts **right**: more NH_3; K unchanged. (b) Shifts **right**: more NH_3 forms; K unchanged. (c) Shifts **right** (4 mol gas → 2 mol gas): more NH_3; K unchanged. (d) Forward reaction is exothermic, so it shifts **left**: less NH_3; **K decreases**. (e) **No shift** and K unchanged; equilibrium is reached faster.

3. Why 450 °C.

> At low temperatures the equilibrium yield is higher, but the **rate** is far too slow, even with a catalyst. 450 °C is a **compromise**: a reasonable yield produced quickly enough. (High pressure and recycling of unreacted gases also improve the overall yield.)

C. Colour and temperature

4. Cooling.

> The forward reaction is endothermic ($\Delta H = +57$ kJ), so cooling favours the **exothermic reverse** reaction. More colourless N_2O_4 forms, and the mixture becomes **paler**.

5. Compression.

> Halving the volume immediately **doubles all concentrations**, so the brown colour first **darkens**. The higher pressure then shifts the equilibrium toward **fewer moles of gas** (2 mol NO_2 → 1 mol N_2O_4), so some NO_2 is used up and the colour becomes slightly **paler**. The change is only partly counteracted, so it stays darker than at the start.

© 2026 Chemistry Clarity · chemistryclarity.com · Free for personal and classroom use. Not for resale.

D. Special cases

6. HI equilibrium.

> There are **2 mol of gas on each side**, so neither direction reduces the number of gas molecules. Changing the pressure changes all concentrations by the same factor, and Q stays equal to K.

7. Adding $CaCO_3$.

> **No shift.** $CaCO_3$ is a pure solid, so it does not appear in $K_c = [CO_2]$; adding more does not change Q.

E. Using Q

8. Q after adding H_2.

> New $[H_2] = 0.0222\ \text{M} + 0.100\ \text{M} = 0.122\ \text{M}$
>
> $$Q = \frac{(0.156\ \text{M})^2}{(0.122\ \text{M})(0.0222\ \text{M})} = 8.99$$
>
> The units of M^2 cancel. $Q = 8.99 < K_c = 49.0$, so the reaction goes **forward** (right).

F. Challenge

9. New equilibrium.

> Let x (in M) be the decrease in $[I_2]$. New equilibrium: $[H_2] = 0.12222\ \text{M} - x$, $[I_2] = 0.02222\ \text{M} - x$, $[HI] = 0.15556\ \text{M} + 2x$. Working in units of M:
>
> $$49.0 = \frac{(0.15556 + 2x)^2}{(0.12222 - x)(0.02222 - x)}$$
>
> Expanding and collecting terms:
>
> $$45.0x^2 - 7.700x + 0.1089 = 0$$
>
> $$x = \frac{7.700 - \sqrt{(7.700)^2 - 4(45.0)(0.1089)}}{2(45.0)} = \frac{7.700 - 6.300}{90.0} = 0.01556$$
>
> (The other root, 0.1556, would make $[I_2]$ negative, so it is rejected.)
>
> $[H_2] = 0.12222\ \text{M} - 0.01556\ \text{M} = \mathbf{0.107\ M}$; $[I_2] = 0.02222\ \text{M} - 0.01556\ \text{M} = \mathbf{0.00667\ M}$; $[HI] = 0.15556\ \text{M} + 0.03111\ \text{M} = \mathbf{0.187\ M}$
>
> Check: $\dfrac{(0.18667)^2}{(0.10667)(0.00667)} = 49.0$ ✓

© 2026 Chemistry Clarity · chemistryclarity.com · Free for personal and classroom use. Not for resale.